

MERGE JOIN AND CONCAT

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In [2]: import pandas as pd
students = pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",
                        sheet_name="students")
marks = pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",
                    sheet_name="marks")
contact = pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",
                    sheet_name="contact")

#1.INNER MERGE
#Return Only Matching Students_IDs from Both DataFrame
inner_df = pd.merge(
    students,
    marks,
    on = "Student_ID",
    how = "inner"
)
print(inner_df.head())
print(inner_df.shape)

#2.Left Merge
#Keeps all records from the Left DataFrame(Students)
left_df = pd.merge(
    students,
    marks,
    on = "Student_ID",
    how = "left"
)
print(left_df.tail())
print(left_df.shape)

#2.Right Merge
##Even if marks are missing ,Students remain.
right_df = pd.merge(
    students,
    marks,
    on = "Student_ID",
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        how = "right"
    )
    print(right_df.tail())
    print(right_df.shape)

#3.Outer Join
#keeps everything from both DataFrame
outer_df = pd.merge(
    students,
    marks,
    on = "Student_ID",
    how = "outer"
)
print(outer_df.tail())
print(outer_df.shape)

#4.MERGE MULTIPLS TABLES
#Combine students + marks + contact
student_complete = pd.merge(
    students,
    marks,
    on = "Student_ID",
    how = "left"
)
student_complete = pd.merge(
    student_complete,
    contact,
    on = "Student_ID",
    how = "left"
)
print(student_complete.head())

#5.Concat Row-Wise
#Split students into two parts
part1 = students.iloc[:25]
part2 = students.iloc[25:]
combined_rows = pd.concat([part1,part2])
print(combined_rows.shape)

#5 Concat Column-Wise
student_basic = students[['Student_ID', 'Name']]
student_marks = marks[['Python']]

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combined_cols = pd.concat(  
    [student_basic, student_marks],  
    axis=1  
)  
print(combined_cols.head())  
  
#6.JOIN - merge data  
#join() works using indexes.  
students_index = students.set_index("Student_ID")  
contact_index = contact.set_index("Student_ID")  
joined_df = students_index.join(contact_index)  
print(joined_df.tail(10))
```

	Student_ID	Name	Department	Maths	Python
0	1001	Student_1	CSE	61	56
1	1002	Student_2	IT	62	57
2	1003	Student_3	ECE	63	58
3	1004	Student_4	EEE	64	59
4	1005	Student_5	MECH	65	60

(45, 5)

	Student_ID	Name	Department	Maths	Python
45	1046	Student_46	CSE	NaN	NaN
46	1047	Student_47	IT	NaN	NaN
47	1048	Student_48	ECE	NaN	NaN
48	1049	Student_49	EEE	NaN	NaN
49	1050	Student_50	MECH	NaN	NaN

(50, 5)

	Student_ID	Name	Department	Maths	Python
45	2001	NaN	NaN	75	80
46	2002	NaN	NaN	75	80
47	2003	NaN	NaN	75	80
48	2004	NaN	NaN	75	80
49	2005	NaN	NaN	75	80

(50, 5)

	Student_ID	Name	Department	Maths	Python
50	2001	NaN	NaN	75.0	80.0
51	2002	NaN	NaN	75.0	80.0
52	2003	NaN	NaN	75.0	80.0
53	2004	NaN	NaN	75.0	80.0
54	2005	NaN	NaN	75.0	80.0

(55, 5)

	Student_ID	Name	Department	Maths	Python	Email
0	1001	Student_1	CSE	61.0	56.0	student1@college.edu
1	1002	Student_2	IT	62.0	57.0	student2@college.edu
2	1003	Student_3	ECE	63.0	58.0	student3@college.edu
3	1004	Student_4	EEE	64.0	59.0	student4@college.edu
4	1005	Student_5	MECH	65.0	60.0	student5@college.edu

(50, 3)

	Student_ID	Name	Python
0	1001	Student_1	56
1	1002	Student_2	57
2	1003	Student_3	58
3	1004	Student_4	59
4	1005	Student_5	60

Name Department

Email

Student_ID			
1041	Student_41	CSE	student41@college.edu
1042	Student_42	IT	student42@college.edu
1043	Student_43	ECE	student43@college.edu
1044	Student_44	EEE	student44@college.edu
1045	Student_45	MECH	student45@college.edu
1046	Student_46	CSE	student46@college.edu
1047	Student_47	IT	student47@college.edu
1048	Student_48	ECE	student48@college.edu
1049	Student_49	EEE	student49@college.edu
1050	Student_50	MECH	student50@college.edu

In []: